

① Solve for x :

$$\tan^{-1} \frac{(x+1)}{(x-1)} + \tan^{-1} \left(\frac{x-1}{x} \right) = \tan^{-1} 7$$

② If for a square matrix A , $A \cdot \text{adj} A = \begin{bmatrix} 2025 & 0 & 0 \\ 0 & 2025 & 0 \\ 0 & 0 & 2025 \end{bmatrix}$ then the value of $|A| + |\text{adj} A|$ is:-

- (A) 1 (B) $2025 + 1$ (C) $2025^2 + 45$ (D) $2025 + 2025^2$

③ $\int \frac{dx}{x^3(1+x^4)^{1/2}} = ?$

④ $\int_0^{2\pi} \csc^7 x \, dx = ?$

⑤ The cost (in rupees) of producing x items in a factory, each day is given by

$$C(x) = 0.00013x^3 + 0.002x^2 + 5x + 2200$$

Find the marginal cost when 150 items are produced.

⑥ If $(x-a)^2 + (y-b)^2 = c^2$ for some $c > 0$, then prove that

$$\frac{\left[1 + \left(\frac{dy}{dx} \right)^2 \right]^{3/2}}{\frac{d^2y}{dx^2}}$$
 is a constant

independent of a and b .

⑦ A point object is placed at the centre of a glass sphere of radius 6cm and refractive index 1.5. The distance of virtual image from the surface of the sphere is.

- Ⓐ 2cm Ⓑ 4cm Ⓒ 6cm Ⓓ 12cm

⑧ A proton is moving with a speed of 2×10^5 m/s enters a uniform magnetic field $B = 1.5$ T. At the entry point velocity vector makes an angle 30° with the magnetic field. Calculate:

- (i) the pitch of helical path described by the charge.
(ii) Kinetic Energy after completing half the circle.

⑨ The half life of a certain radioactive material against α -decay is 100 days. After how much time will the undecayed fraction of the material be 6.25%?

⑩ In a given coil of self inductance of 5 mH, current changes from 4A to 1A in 30 ms. Calculate the emf induced in the coil.

⑪ Which of the following contains only β -D-glucose as its monosaccharide unit?
a) sucrose b) cellulose c) starch d) Maltose

⑫ Which one of the following sets correctly represents the increase in the paramagnetic property of the ions?

- a) $Ti^{+3} < Fe^{+2} < Cr^{+3} < Mn^{+2}$
b) $Ti^{+3} < Mn^{+2} < Fe^{+2} < Cr^{+3}$
c) $Mn^{+2} < Fe^{+2} < Cr^{+3} < Ti^{+3}$
d) $Ti^{+3} < Cr^{+3} < Fe^{+2} < Mn^{+2}$

⑬ Chloroform (B.P $61.2^{\circ}C$) and acetone (B.P $56^{\circ}C$) are mixed to form an azeotrope.

The mole fraction of acetone in this mixture is 0.339. Predict whether the boiling point of this azeotrope will be

(i) $60^{\circ}C$ (ii) $64.5^{\circ}C$ (iii) $54^{\circ}C$ • Defend your answer with reason.

⑭ Write the IUPAC name of the following complex: $K[Cr(H_2O)_2(C_2O_4)_2]H_2O$

⑮ Name the metal present in the complex compound of

- a) Haemoglobin b) Vitamin B-12